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ubuntu@vm1:~$ python3 -m venv ~/lcd
ubuntu@vm1:~$ source ~/lcd/bin/activate
(lcd) ubuntu@vm1:~$ pip install 'llvmlite==0.49.*'
Collecting llvmlite==0.49.*
  Downloading
llvmlite-0.49.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata
(5.0 kB)
Downloading llvmlite-0.49.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl
(59.9 MB)

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59.9/59.9 MB 10.1 MB/s eta 0:00:00
Installing collected packages: llvmlite
Successfully installed llvmlite-0.49.0
(lcd) ubuntu@vm1:~$ python3 -c "import llvmlite.binding as b; print(b.llvm_version_info)"
(22, 1, 0)
(lcd) ubuntu@vm1:~$ touch test.ll
(lcd) ubuntu@vm1:~$ nano test.ll
(lcd) ubuntu@vm1:~$ nano test.ll
(lcd) ubuntu@vm1:~$ nano test.ll
(lcd) ubuntu@vm1:~$ llc
^C
(lcd) ubuntu@vm1:~$ llc test.ll
llc: error: llc: test.ll:4:2: error: multiple definition of local value named 'sum'
    %sum = add i32 %sum, 1
    ^
(lcd) ubuntu@vm1:~$ nano test.ll
(lcd) ubuntu@vm1:~$ nano test.ll
(lcd) ubuntu@vm1:~$
(lcd) ubuntu@vm1:~$ llc -filetype=obj test.ll -o test.o
(lcd) ubuntu@vm1:~$ clang test.o -o test
(lcd) ubuntu@vm1:~$ ./test; echo $?
8

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(lcd) ubuntu@vm1:~$ llc -filetype=asm test.ll -o -
    .text
    .file    "test.ll"
    .globl  main                # -- Begin function main
    .p2align    4, 0x90
    .type    main,@function
main:                                # @main
    .cfi_startproc
# %bb.0:                                # %entry
    movl    $8, %eax
    retq
.Lfunc_end0:

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        .size    main, .Lfunc_end0-main
        .cfi_endproc

        # -- End function

        .section      ".note.GNU-stack","",@progbits
(lcd) ubuntu@vm1:~$ file test.o test
test.o: ELF 64-bit LSB relocatable, x86-64, version 1 (SYSV), not stripped
test:  ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter
/lib64/ld-linux-x86-64.so.2, BuildID[sha1]=db85bda1d681e7701227a1ee138c5eaaefe7c0b3, for
GNU/Linux 3.2.0, not stripped
(lcd) ubuntu@vm1:~$ readelf -h test.o | grep -E 'Class|Type|Machine'
Class:                ELF64
Type:                REL (Relocatable file)
Machine:             Advanced Micro Devices X86-64
(lcd) ubuntu@vm1:~$ nm test.o
0000000000000000 T main
(lcd) ubuntu@vm1:~$ nm test | grep -E ' main| _start'
0000000000001040 T _start
0000000000001130 T main
(lcd) ubuntu@vm1:~$ objdump -d test.o
test.o:  file format elf64-x86-64
Disassembly of section .text:
0000000000000000 <main>:
   0:  b8 08 00 00 00      mov    $0x8,%eax
   5:  c3                  ret
(lcd) ubuntu@vm1:~$

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